CHAPTER 10: Computer Networks

Question 1: What is ping, and why does "no reply" not always mean a website is down?

What is ping? A network diagnostic command line utility that tests whether a specific host is reachable across an IP network. It works by sending ICMP Echo Request (Internet Control Message Protocol) packets to the target IP address and listening for ICMP Echo Reply packets, while measuring round-trip time (RTT) and packet loss.

Why "no reply" doesn't mean it's down: Many modern websites, cloud servers (e.g., AWS EC2), and enterprise firewalls intentionally drop or block ICMP packets to prevent denial-of-service (DDoS) ping floods and hide internal network topology. The web server might be running an active, healthy HTTPS web service on port 443 while remaining completely silent to ping requests.

Question 2: How the Internet Functions with >20 Billion Devices on 4.3 Billion IPv4 Addresses (Excluding IPv6)

The fundamental technology that prevented IPv4 exhaustion is NAT (Network Address Translation), specifically NAPT / PAT (Port Address Translation), coupled with Private IP Address Spaces (RFC 1918):

Private vs. Public Addresses: Devices inside private local networks (homes, university hostels, corporate offices) are assigned non-routable private IP addresses (such as 192.168.x.x or 10.x.x.x). Millions of different private networks worldwide reuse these exact same internal IP ranges simultaneously without conflict.

Port Address Translation (PAT): ~~tdehr~~ When dozens or hundreds of ~~devicces~~ devices on a local Wi-Fi router access the public internet, the gateway router translates all their private IP addresses into a single public routable IPv4 address. The router differentiates internal devices by mapping each outbound TCP/UDP connection to a unique source port number.

Carrier-Grade NAT (CGNAT): Internet Service Providers (ISPs) take this a step further by placing thousands of households behind a shared tier of ISP-level public IPs.

Application-Layer Sharing (Reverse ~~Proxies~~ Proxies & CDNs): Multiple domain names share identical public IP addresses using Server Name Indication (SNI) and HTTP/Host headers, allowing CDNs (like Cloudflare) to serve millions of websites behind a small pool of IP addresses.

Together, these technologies allow tens of billions of connected devices to operate seamlessly over the limited 32-bit IPv4 address space.